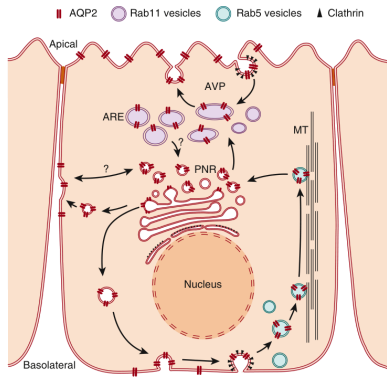


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Fig_10_10

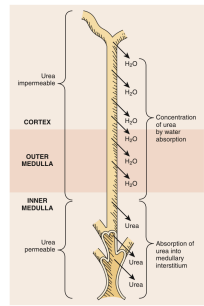


Fig. 10.13 Schematic representation of the mammalian collecting duct system showing principal sites of water absorption and urea absorption. Water is absorbed in the early part of the collecting duct system, driven by an osmotic gradient. Since urea permeabilities of cortical, outer medullary, and initial inner medullary collecting duct are low, the water absorption concentrates urea in the lumen of these segments. When the tubule fluid reaches the terminal inner medullary collecting duct, which is highly permeable to urea, urea rapidly exits from the lumen. This urea is trapped in the inner medulla because of countercurrent exchange.

Fig_10_13

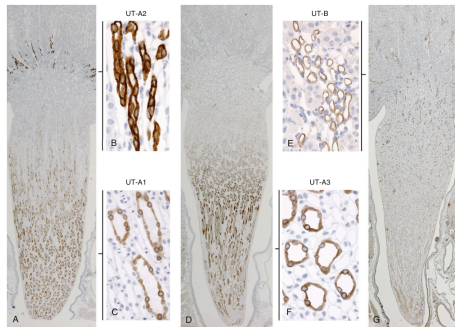
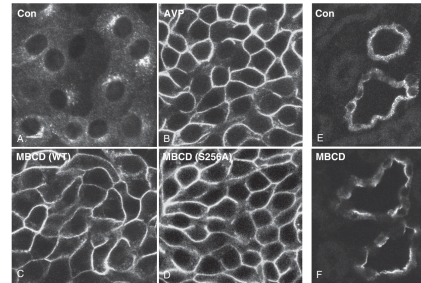


Fig. 10.16 Localization of urea transporters. UT-A1 is localized to the terminal portion of the inner medullary collecting duct, whereas UT-A2 and UT-B are found descending in the descending vasa recta (A). The thin stripe of outer medulla (A), the thin stripe of outer medulla (B), and the UT-A2 (C) and UT-A1 (C) are predominantly intracellular. UT-A3 is localized to the terminal portion of the inner medullary collecting duct (D) and is both intracellular and in the basolateral membrane domains (F). UT-B is expressed in the descending vasa recta (G), where it is localized to the basolateral and apical regions (E). (Adapted from Fenton RA, Knepper MA. Urea and renal function in the 21st century: insights from knockout mice. *Am Soc Nephrol*. 2007;18: 679–688.)

Fig_10_16

[illegible]

Fig_10_11

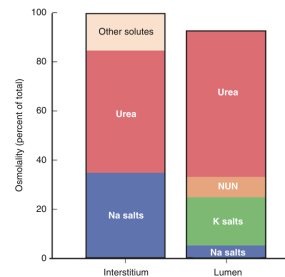
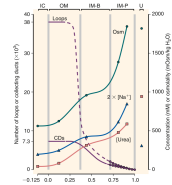


Fig. 10.14 Solutes that account for osmolality of medullary interstitium and tubule fluid in the inner medullary collecting duct during antidiuresis in rats. Urea nearly equilibrates across the inner medullary collecting duct epithelium because of rapid facilitated urea transport. Although the osmolality of the fluid in the two spaces are nearly equal, the nonurea solutes can differ considerably between the two compartments. Typical values in untreated rats are presented. Values can differ considerably in other species and in the same species with different diets. *NUN*, Nonurea nitrogen.

Fig_10_14

[illegible]

Fig_10_17

		Long loop		Short loop		Collecting duct	
Medullary rays			PST (1.5)		CTAL (1.5)		CCD (0.4)
Outer medulla	Outer stripe	PST (2.1)	MTAL (1.4)	PST (2.1)	MTAL (1.4)		
	Inner stripe	LDL _{CM} (1.5)	MTAL (0.7)	SDL (7.5)	MTAL (0.9)		OMCD (3.5)
Inner medulla							IMCD, (3.4)
		LDL _{IM} (13.5)	ATL (19)				

Fig. 10.12 Urea permeabilities of mammalian renal tubule segments. The width of each segment in the diagram is distorted to be proportional to the urea permeability of that segment. Numbers in parentheses are measured values for the permeability coefficient ($\times 10^{-5}$ cm/sec). Values are from isolated perfused tubule studies. *ATL*, Ascending thin limb; *MTAL*, cortical thick ascending limb; *IMTL*, inner medullary collecting duct; *IMCD*, terminal inner medullary collecting duct; *LDL_{out}*, thin descending limb of long-looped nephron in outer medulla; *MTAL*, medullary thick ascending limb; *OMCD*, outer medullary collecting duct; *PST*, proximal straight tubule; *SDL*, thin descending limb of short-looped nephron.

Fig_10_12

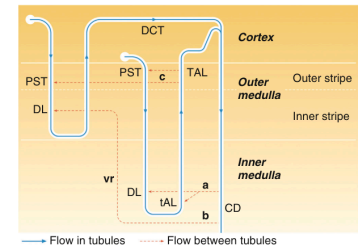


Fig. 10.15 Pathways of urea recycling in renal medulla. Solid blue lines represent a short-looped nephron (left) and a long-looped nephron (right). Transfer of urea between nephron segments is indicated by dashed red arrows labeled a, b, and c corresponding to recycling pathways described in the text. CD, Collecting duct; DCT, distal convoluted tubule; DL, descending limb; PST, proximal straight tubule; TAL, thin ascending limb; TAL, thick ascending limb; vr, vasa recta. (Knepper MA, Ruch-Panel F. Pathways of urea transport in the mammalian kidney. *Kidney Int.* 1987;31:629-633.)

Fig_10_15

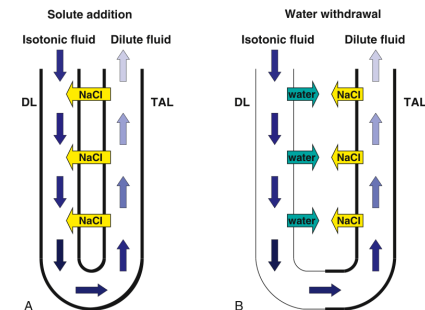
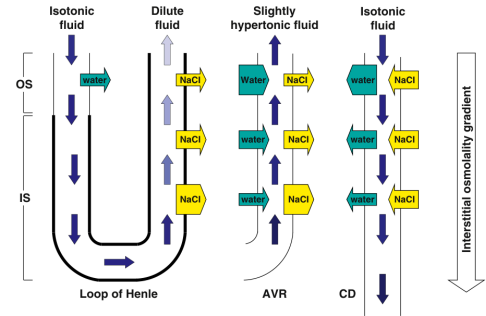


Fig. 10.18 Theoretical mechanisms of countercurrent multiplication in the loop of Henle. (A) Countercurrent multiplication by means of NaCl transfer from an ascending flow to a descending flow. (B) Countercurrent multiplication by means of water withdrawal from a descending flow. NaCl transport from the ascending flow into the interstitium raises interstitial osmolality; this results in passive water transport from the descending flow, which has lower osmolality than the interstitium. In both panels, tubular fluid flow direction is indicated by blue arrows; increasing osmolality is indicated by darkening shades of blue. DL, Descending limb of Henle loop; TAL, thick ascending limb of Henle loop. Thick black lines indicate that a tubule is impermeable to water; thin lines indicate high permeability to water.

Fig_10_18

Fig. 10.19 Outer medullary concentrating mechanism based on NaCl addition to the interstitium but without water absorption from descending limbs of short loops. Arrows indicate water (teal) and NaCl (yellow) transepithelial transport; arrow widths suggest relative transport magnitudes. Isotonic fluid is considered to have the same osmolality as blood plasma. Flow entering the AVR is assumed to arise from a descending vasa rectum that is in, or near, a vascular bundle. Outflow from the collecting duct enters the inner medullary collecting duct. Tubular fluid flow direction is indicated by blue arrows; increasing osmolality is indicated by darkening shades of blue. Thick black lines indicate that a tubule is impermeable to water; thin lines indicate high permeability to water. AVR, Ascending vasa recta; CD, collecting duct; IS, inner stripe; OS, outer stripe.



Fig_10_19